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SECTION : G

Lab 1: Requirements Engineering & UML Use-Case Modelling

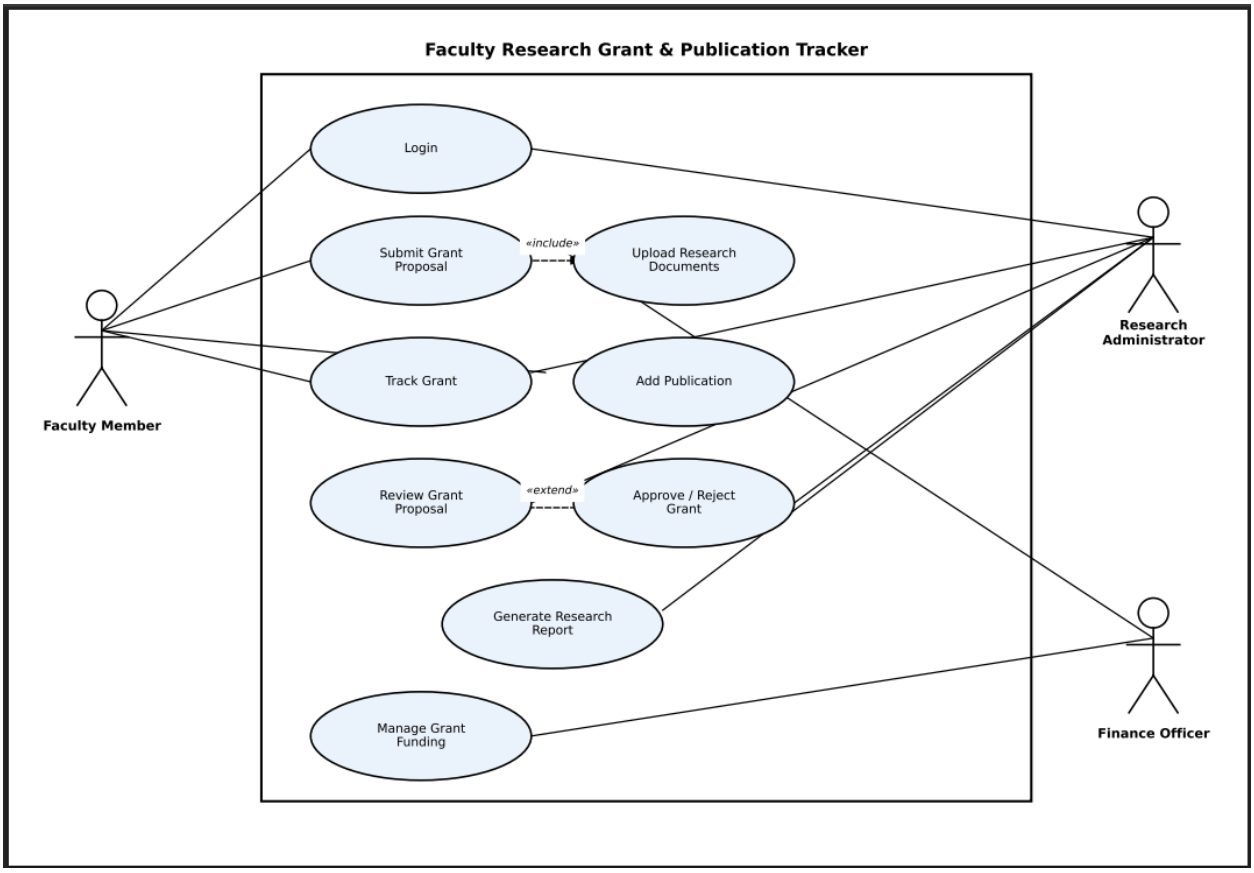
Problem: Faculty Research Grant & Publication Tracker

Deliverable 1 — SRS / Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow a Faculty Member to create and submit a research grant proposal, capturing title, description, requested funding amount, and proposed duration.	High	Given all mandatory fields are filled, when the Faculty Member clicks Submit, then the system stores the proposal with status "Submitted" and displays a unique Proposal ID within 3 seconds. If a mandatory field is empty, submission is blocked and an inline error is shown.	Faculty members need a structured, centralized way to submit proposals instead of email or paper forms, as required by the problem statement.	Core entry point for the grant workflow.
FR-002	Functional	The system shall allow a Faculty Member to upload supporting documents (e.g., detailed proposal document, budget sheet) as part of a grant proposal.	High	Given a proposal in progress, when the Faculty Member attaches a file of an accepted type (PDF/DOCX, max 10 MB), then the file is listed under the proposal. Uploading an unsupported file type is rejected with an error message.	Grant proposals require supporting evidence; the problem statement explicitly calls for document upload capability.	Linked to FR-001 via the Submit Grant Proposal use case.
FR-003	Functional	The system shall allow a Research Administrator to review a submitted grant proposal and record a decision to approve or reject it,	High	Given a proposal with status "Submitted", when the Research Administrator selects Approve or Reject, then the proposal status updates accordingly and	Administrators must be able to formally evaluate proposals, as stated in the problem statement.	Status change must be visible to the Faculty Member (see FR-004).

		with an optional comment.		the decision timestamp is recorded. A decision cannot be recorded twice on the same proposal.		
FR-004	Functional	The system shall allow a Faculty Member to view the real-time status of their submitted grant proposals (Submitted, Under Review, Approved, or Rejected).	Medium	Given a logged-in Faculty Member, when they open "Track Grant", then all their proposals are listed with the current status matching the latest administrator decision.	Faculty members need timely access to their grant records, as required by the problem statement.	Depends on status updates made under FR-003.
FR-005	Functional	The system shall allow a Research Administrator to generate a research report summarizing grant proposals and publications within a specified date range.	Medium	Given a selected start and end date, when the Research Administrator clicks Generate Report, then a report listing all grants and publications in that range is produced within 5 seconds, or a "No records found" message is shown if none exist.	Administrators need to monitor grant progress and produce summaries, as required by the problem statement.	Report should be exportable (e.g., PDF/CSV) in a future iteration.
NFR-001	Nonfunctional (Security)	The system shall restrict access to grant and publication records based on the logged-in user's role (Faculty Member, Research Administrator, or Finance Officer).	High	Given a user logged in as Faculty Member, when they attempt to access administrator-only functions (e.g., Approve/Reject Grant) via direct URL access, then the system denies access and displays an "Unauthorized" message.	The problem statement requires protecting research data from unauthorized users.	To be implemented via role-based access control (RBAC).
NFR-002	Nonfunctional (Performance)	The system shall load the grant status and dashboard pages within 3 seconds under normal load conditions of up to 100 concurrent users.	Medium	Given 100 simulated concurrent users, when a user requests the dashboard page, then the page fully renders within 3 seconds for at least 95% of requests, measured using a load-testing tool.	The problem statement requires timely access to records; slow response undermines usability.	To be validated during performance testing before deployment.

Deliverable 2 — UML Use-Case Diagram



Deliverable 3 — Use-Case Flow

Use Case Name	Submit Grant Proposal
Primary Actor	Faculty Member
Secondary Actors	None (system performs storage and validation automatically)
Related Use Case	Includes: Upload Research Documents («include»)

Preconditions

1. The Faculty Member has a valid registered account and is logged into the system.

2. The Faculty Member has the proposal details (title, description, requested funding amount, duration) and supporting documents ready.

Main Success Scenario

1. The Faculty Member selects the "Submit Grant Proposal" option from the dashboard.

2. The system displays a grant proposal form with fields for title, description, requested funding amount, and duration.

3. The Faculty Member enters the required proposal details into the form.

4. The Faculty Member uploads the required supporting documents (this step invokes the included use case "Upload Research Documents").

5. The Faculty Member clicks the "Submit" button.

6. The system validates that all mandatory fields are completed and at least one required document is attached.

7. The system saves the proposal with status "Submitted", generates a unique Proposal ID, and timestamps the submission.

8. The system displays a confirmation message to the Faculty Member containing the Proposal ID and submission status.

Alternate Flow A1 — Required Document Missing

Trigger: Occurs at step 6 of the Main Success Scenario, when the Faculty Member attempts to submit the proposal without attaching a mandatory supporting document.

1. The system detects that a required document has not been uploaded.

2. The system halts the submission and displays a validation error message: "Please upload the required supporting document(s) before submitting."

3. The Faculty Member uploads the missing document(s).

4. The Faculty Member clicks "Submit" again.

5. The flow resumes at step 6 of the Main Success Scenario.

Alternate Flow A2 — Mandatory Field Left Empty

Trigger: Occurs at step 6, when one or more mandatory text fields (title, description, funding amount, duration) are empty.

1. The system identifies the missing field(s) and highlights them in red.

2. The system displays the message: "Please fill in all required fields."

3. The Faculty Member completes the missing field(s) and clicks "Submit" again.

4. The flow resumes at step 6 of the Main Success Scenario.

Postconditions

1. A new grant proposal record exists in the system with status "Submitted", a unique Proposal ID, and all attached documents linked to it.

2. The proposal becomes visible to the Faculty Member under "Track Grant" and to Research Administrators under "Review Grant Proposal".